A Leakage-Controlled Reproduction of Dream2Learn

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***Abstract*—Dream2Learn (D2L) reports that optimized generative "dreams" improve rehearsal-based continual learning. We independently reimplemented its central algorithm on Split Mini-ImageNet, replacing the externally pretrained diffusion model with a conditional variational autoencoder (CVAE) trained from random initialization on the Continual Learning training data. Using a shared improved-CVAE initialization, we compared replay alone, random latent dreams, and directed latent dreams. Directed dreaming reached 19.76% final average accuracy, below replay by 2.30 percentage points and random dreams by 3.31 points. It nevertheless increased full-head forward transfer to +4.44 points, compared with -1.08 for replay. The primary final-accuracy claim did not reproduce under this leakage-controlled adaptation. We report the protocol changes, ambiguities in the original description, and a directed-versus-random control absent from the paper.**

***Index Terms*—continual learning, generative replay, variational autoencoder, reproducibility**

I. Claim Reproduced

[1] propose preparing a continual learner for unseen concepts by generating "dreams" from previously learned classes. We reproduce their central claim from Section 4.3: D2L is an "orthogonal augmentation to rehearsal" that improves class-incremental final average accuracy (FAA). FAA is measured after the final task on a test set containing every class. The paper supports the claim with Table 1 and reports forward transfer (FWT) in Table 2, both on PDF p. 8.

The central comparison in this reproduction is Experience Replay (ER) on Split Mini-ImageNet with a 2,000-example buffer. Adding D2L raised FAA by 3.27 percentage points and FWT by 2.91 points (Table I). The paper also evaluates other rehearsal methods, datasets, and a 5,000-example buffer, but ER is the clearest reference because our baseline is standard replay rather than DER++ or ER-ACE.

The authors used a randomly initialized ResNet-18 classifier and a frozen pretrained Stable Diffusion generator. They optimized prompts in the generator's text space, stopped trajectories with a learned oracle, generated images from the retained prompts, and trained the continual learner on those images. Their figures are means and standard deviations over five runs.

TABLE I
Original ER results on Split Mini-ImageNet with replay capacity 2,000, summarized from Tables 1 and 2 of [1].

Original condition	FAA	FWT
ER	27.91 ± 3.49	-2.58
ER + D2L	31.18 ± 2.74	$+0.33$
D2L effect	$+3.27$	$+2.91$

TABLE II
Material differences from the original experiment.

Component	Dream2Learn	This reproduction
Code	Authors' method	Independent reimplementation
Generator	Pretrained Stable Diffusion	CVAE trained only on CL training images
Dream domain	Text prompts and images	Additive latent offsets
Classifier	ResNet-18	CVAE encoder and linear head
Stopping	Learned oracle	Fixed observed-data heuristics

II. Reimplementation and Scope

We did not use the authors' code. We independently reimplemented the D2L algorithm, defined as "Algorithm 1" in the paper, with some modifications listed in Table II. This is deliberately not an exact replication. It is a leakage-controlled reproduction of the mechanism claim: can directed dreaming improve rehearsal when the generator has not learned semantic information from external data?

[1]'s use of a pretrained diffusion model raises concerns about data leakage. The diffusion model was pretrained on external image data, potentially including concepts related to future unseen classes. The authors acknowledge the existence of "feature-level proximity" between the pretrained model's representations and future unseen classes. However, they consider this the intended outcome of D2L, stating: "[The dreams'] non-uniform proximity to future classes instead supports the intended role of D2L, namely shaping reusable representations without revealing future labels." In order to isolate the effect of the dream procedure itself, we remove the pretrained model from our reproduction. We therefore replaced the pretrained diffusion with a CVAE trained only on training images. Dreams are transformations in its latent space and train the classifier outputs directly. *Note: Decoded images are diagnostics rather than training examples.*

III. Experimental Protocol

We use the paper's high-level Split Mini-ImageNet stream: 50 classes in the base task followed by ten tasks of five classes. The replay capacity is 2,000. All final conditions are initialized from the same 100-epoch CVAE base checkpoint.

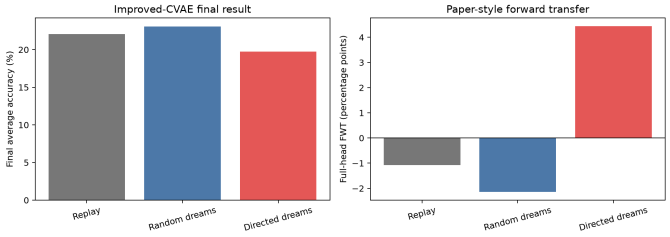


Fig. 1. Seed-42 final average accuracy and full-head FWT after replacing the pretrained generator with the improved CVAE.

We compare three conditions that differ only in their treatment of dreams. *Replay only* trains on current-task images and a bounded memory of earlier images. *Random dreams* adds random latent offsets to the same training process. *Directed dreams* instead optimizes those offsets toward learned guide classes using classifier cross-entropy. Comparing directed with replay tests the overall augmentation, whereas comparing directed with random dreams isolates whether the optimization direction itself helps.

The primary outcome is the final average accuracy (FAA) on the test set after the last task. We also report sample-weighted accuracy and average forgetting. For FWT, the learner is evaluated on each incoming task immediately before training on it. We subtract the corresponding accuracy of a separately initialized random model and average across incoming tasks. Our reported *full-head FWT* makes predictions across all 100 classifier outputs and is therefore closest to the paper's class-incremental metric.

IV. Results

Directed dreaming reached 19.76% FAA, which is -2.30 percentage points below replay and -3.31 points below random dreams (Table III). Random dreams were instead 1.00 point above replay. The directed-versus-random difference is a number the paper does not report and is the sharper mechanism test: the random condition receives the same auxiliary latent capacity and output-reuse lifecycle without directed optimization. On the primary metric, optimizing dream direction was therefore harmful in this adapted setting.

The original paper's closest ER comparison gained 3.27 points, whereas our directed condition lost 2.30 points against replay. Absolute accuracies are not like-for-like: replacing ResNet-18 and Stable Diffusion changes both model capacity and prior knowledge. The more defensible comparison is therefore the sign of the within-protocol augmentation effect. The original reports a positive effect, while the leakage-controlled latent method produces a negative one.

FWT gives a different result. Replay and random dreams have negative FWT (-1.08 and -2.14 points), whereas directed dreams reach +4.44 points. Directed optimization therefore improves FWT by 5.52 points over replay and 6.58 over random dreams. This directionally agrees with the paper's ER result, where D2L changed FWT from -2.58 to $+0.33$. However, the improved anticipation does not survive as higher final accuracy. Directed dreams also have the most forgetting at 71.99 points, compared with 70.22 for replay and 70.18 for random dreams. Their sample-weighted final accuracy is 14.17%, versus 13.88% and 19.12%, respectively. The disagreement between transfer and retention is therefore not an effect of reporting only task-average accuracy.

V. What Did Not Reproduce

The main failed reproduction is the paper's final-accuracy improvement. Directionally optimized dreams neither improve on Experience Replay nor beat the random-dream control. Random dreams perform best despite having no class-directed optimization, suggesting that auxiliary latent training or output reuse explains more of the final benefit in our setting. The FWT result does partially reproduce the anticipated-transfer effect, but the greater forgetting shows why this is insufficient evidence for the broader claim.

Several missing details prevented a strict reconstruction. The paper does not fully specify class orders, image preprocessing and augmentation, replay batch composition, or the main experiment's number of generated images per dream class. This lack of detail limited our ability to exactly replicate the original experimental setup.

We made these choices explicit rather than silently treating them as equivalent to the paper. Training images use random resized crops and horizontal flips, while validation and test images use deterministic resize and centre crop. The classifier and CVAE are optimized jointly with AdamW at learning rate 0.0003; the paper reports SGD at 0.03 for its classifier but does not enumerate all optimizer settings needed for an exact reconstruction. The selected CVAE also receives 100 base epochs rather than the paper's 10 classifier epochs. These changes improve the learned representation but alter both compute and optimization scale, so the absolute FAA values cannot establish a direct performance shortfall.

The final improved-CVAE comparison is one seed, whereas the paper reports five runs. We therefore report exact paired differences from the shared seed-42 initialization but do not attach a confidence interval or claim statistical significance. Sharing the base checkpoint removes initialization noise between our three conditions, making their within-seed comparison more controlled, but it cannot measure sensitivity to class order or training randomness. Replicating the balanced model across additional seeds would be the first extension if time permitted.

The largest intentional change is also the largest limitation. The CVAE removes pretrained semantic information but has weaker generative capacity than Stable Diffusion. Consequently, these results do not prove that Dream2Learn's gains are caused by leakage. They show that the directed mechanism did not transfer into improved retention in this controlled latent implementation. The pretrained generator's semantic prior, its image manifold, or another changed component may be necessary for the reported FAA gains.

VI. Conclusion

We attempted to reproduce Dream2Learn's claim that generative dreaming is an orthogonal augmentation that improves

TABLE III

ORIGINAL AND REPRODUCED SPLIT MINI-IMAGENET RESULTS WITH REPLAY BUFFER 2,000. ABSOLUTE VALUES ARE NOT LIKE-FOR-LIKE BECAUSE THE GENERATOR AND CLASSIFIER DIFFER.

Condition	FAA	Sample-weighted FAA	Average forgetting	Full-head FWT
Original ER	27.91 ± 3.49	—	—	-2.58
Original ER + D2L	31.18 ± 2.74	—	—	$+0.33$
Replay only	22.06%	13.88%	70.22	-1.08
Random dreams	23.06%	19.12%	70.18	-2.14
Directed dreams	19.76%	14.17%	71.99	+4.44

rehearsal. With a CVAE trained only on observed stream data, directed latent dreams achieved 19.76% FAA, -2.30 points relative to replay and -3.31 points relative to random dreams. The primary final-accuracy claim therefore did not reproduce. Directed dreams did, nevertheless, change full-head FWT from -1.08 to +4.44 points relative to replay, partially reproducing the transfer claim without improving long-term retention. The result narrows rather than refutes Dream2Learn: its FAA gain may depend on the pretrained generator's semantic prior or image manifold. It also demonstrates why random-dream controls and both transfer and retention metrics are needed to distinguish directed dreaming from generic auxiliary augmentation.

REFERENCES

[1] S. Calcagno, M. Pennisi, F. Proietto Salanitri, A. Sorrenti, S. Palazzo, C. Spampinato, and G. Bellitto, "Dream2Learn: Structured Generative Dreaming for Continual Learning," *International Journal of Computer Vision*, vol. 134, article 388, 2026, doi: 10.1007/s11263-026-02958-1.